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PERFORMANCE SPECIFICATION

FILTER-SEPARATOR, LIQUID FUEL SKID-MOUNTED, 600 GALLONS PER MINUTE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification outlines the minimum requirements for a horizontal or vertical, liquid fuel separator with a maximum flow rate of 600 gallons per minute (gpm) ±10%. This FFU-15/E filter separator is intended for use with the A/E32R-14 fuel-servicing unit for refueling aircraft.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and Standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2):

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be sent to: San Antonio Air Logistics Center, 485 Quentin Roosevelt Rd., Kelly AFB, TX 78241-6425 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4930

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SPECIFICATIONS

FEDERAL

A-A-52592 - Pipe Fittings: One or More Ends Grooved
A-A-52598 - Couplings, Clamp, Pipe; with Bolts and Synthetic-Rubber Gaskets for Grooved-End Pipe
A-A-55804 - Rods, Ground (with Attachments)
A-A-59326 - Coupling Halves, Quick Disconnect, Cam-Locking Type

DEPARTMENT OF DEFENSE

MIL-DTL-5624 - Turbine Fuel, Aviation, Grades JP-4, and JP-5/JP-8 ST
MIL-PRF-25017 - Inhibitor, Corrosion/Lubricity Improver, Fuel Soluble
MIL-DTL-83420 - Wire Rope, Flexible, for Aircraft Control

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement

DEPARTMENT OF DEFENSE

MIL-STD-130 - Identification Marking of U.S. Military Property

(Unless otherwise indicated, copies of the above specifications, standards, and handbook are available from the Defense Automated Printing Service (DAPS), 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094, phone 215/697-2197).

2.3 Non-Government documents. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of DoDISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN PETROLEUM INSTITUTE (API)

API Bulletin 1581 - Aviation Jet Fuel Filter/Separators, Specification and Qualification Procedures for (DoD adopted).

(Application for copies should be addressed to the American Petroleum Institute, 1220 L Street, NW, Washington, D.C. 20005, phone 202/682-8000 or FAX 202/682-8030.)

AMERICAN WELDING SOCIETY (AWS)

AWS B2.1-84 - Procedure, Welding, and Performance Qualification, Specification for (DoD adopted)
AWS D1.2-83 - Aluminum, Structural Welding Code, Standard for (DoD adopted)

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, Miami FL 33126, phone 305/443-9353 ext.297 or FAX 305/443-5951.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Sections VIII & IX, D1B - Boiler and Pressure Vessel Code, Rules for Construction of Unfired Pressure Vessels (DoD adopted)

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 E. 47th St., New York, NY 10017, phone 212/705-8500 or FAX 212/705-7739.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. PERFORMANCE REQUIREMENTS

3.1 First Article. When specified (see 6.2.4) a sample shall be subject to first article inspection in accordance with 4.2.

3.2 Ease of use. The filter separator shall be designed to insure ease and rapid installation of equipment, ease of use, and ease of general maintenance and repair.

3.3 Performance requirements. The filter separator shall be capable of meeting the performance requirements outlined in the 3rd Edition of American Petroleum Institute's Bulletin 1581, "Specification and Qualification Procedures for Aviation Jet Fuel Filter Separators", for Group II, Class B equipment at a design flow rate of 600 gpm (±10 gpm).

3.3.1 Filter separator. The filter separator shall be designed and constructed so that parts will not work loose and capable of withstanding the stresses, jars, vibrations and other conditions incidental to shipping, storage, and service. All components shall be easily removed, serviced and operated. Any design which renders operation of the filter separator as unduly difficult under field conditions will not be acceptable. The filter separator, coalescers, and separator screens shall be identified with the manufacturer's part number.

3.3.2 Components. To meet user safety and maintenance requirements, the filter separator shall consist of the following major components: vessel, vessel cover, base or skid frame, coalescer and separators elements, flow control valve, float control assembly, tie downs, fittings, connections and miscellaneous openings, spider assembly, differential pressure gauge, and ground rod.

3.3.2.1 Materials. All components and materials (to include gaskets, fittings, valves, adhesives, etc.) that normally come into contact with the fuel during operation shall be proven fuel resistant materials and shall not degrade or be affected by fuel.

3.3.3 Accessories. The filter separator accessories shall consist of a direct read, piston-type differential gauge that is capable of measuring differential pressure across both coalescers and separators and have an automatic air eliminator with a check valve feature. It shall be designed to vent the vessel for element change and include an adjustable pressure relief valve, a manual ball valve for draining the vessel, and a flow control/water slug valve.

3.3.4 Flow. The maximum flow through the filter separator shall be limited to 600 gpm.

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3.3.5 Air in vessel. The filter separator shall be capable of eliminating air in the vessel, as well as air introduced into the vessel, out through the air eliminator, see 4.5.3.

3.3.6 Pressure loss. Any pressure loss shall not exceed 20 psi from the inlet to the outlet connection at a flow rate of 600 gpm using 0.76 specific gravity grade JP-4, JP-5/JP-8 ST fuel in accordance with MIL-DTL-5624 when the elements are in new condition.

3.4 Protective treatment. Materials used in the construction of the filter separator shall be protected against corrosion and deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with use, age, or extremes of climatic and environmental conditions shall be avoided.

3.5 Welding. Welding shall be in accordance with ASME “Boiler and Pressure Vessel Code, Section 9”, AWS B2.1-84; and AWS D1.2-83.

3.6 Assembly. Each filter separator unit shall be delivered fully assembled and shall be ready for use. The functioning of all components shall be tested prior to shipment and no assembly or field adjustment of valves or components shall be required before use.

3.6.1 Vessel. The vessel shall be designed and fabricated in accordance with AWS D1.2, Section 9 and qualified to the API Bulletin 1581, Group II, Class B. Swing type bolts shall be attached to the vessel for safety reasons to retain the cover in a lock-tight position.

3.6.2 Vessel cover. A removable vessel cover shall be provided and shall permit element replacement. The cover shall be provided with handles to facilitate its removal and replacement, an O-ring type gasket to ensure the filter separator seals properly and performance is met during hydrostatic pressure test (see 4.5.2).

3.6.3 Coalescer and separator elements. Coalescer elements are to be threaded base and 6-inch diameter. Separators are to be constructed to allow screen flow from outside-in to ensure performance and endurance.

3.6.4 Working pressure. The vessel shall be designed for a minimum working pressure of 150 psi and shall be constructed and labeled in accordance with Section 9 of the ASME Code for Unfired Pressure Vessels.

3.6.4.1 Inlet and outlet connections. The inlet and outlet connections shall be 4 inch nominal pipe size.

3.6.4.2 Flow control valve. A filter separator flow control value shall be provided with the vessel and shall match the outlet connection flange face.

3.6.4.3 Inlet pipe. The inlet pipe shall have a check valve to ensure performance.

3.6.4.4 Outlet face. The outlet pipe shall be equipped with a rate of flow control/water slug valve.

3.6.4.5 Float control assembly. The float control assembly and its integral float control tester shall provide a means to mechanically adjust the float to verify operation of the water slug valve and integrity of the float valve.

3.6.5 Drain. The vessel shall be equipped with a 0.750 inch manual ball valve water and fuel drain.

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3.6.6 Pressure gauge. A direct-reading, piston type, differential pressure gauge that measures the differential pressure across coalescers and separators shall be installed to ensure performance and endurance.

3.6.7 Air eliminator. An automatic air eliminator shall be installed on top of the vessel and a 0.750 inch angle pattern pressure relief valve shall be provided to ensure performance.

3.6.8 Connections. Sampling connections shall be provided at the inlet and outlet connections.

3.6.9 Base or skid. A base shall be provided for supporting, mounting, and retaining the components specified in 3.6. The base shall not exceed 66 inches long, 42 inches wide, and 60 inches high (for vertical filter separator vessels) and 84 inches long, 48 inches wide, and 47 inches high (for horizontal filter separator vessels). The base or skid shall have sufficient structural integrity to prevent damaging stresses and forces from being transmitted to the components. It shall have tie and lift provisions to permit the unit to be lifted by a sling or a forklift truck, or situated on rough terrain when in a filled condition. Unless otherwise specified, components shall be retained by removable fasteners such as nuts and bolts except the vessel may be retained to the base by welding. When nuts and bolts are used, the nuts shall be of self-locking type. The ends of the base shall be sloped at approximately 30° and shall have a rounded end (approximately 1 inch radius) where the slope joins the top of the base.

3.6.10. Weight. The filter separator shall weigh less than 900 pounds in a drained condition.

3.6.11 Tie downs. The filter separator shall be provided with sufficient tie down points attached so the assembly can be restrained in all directions during air transport. The tie downs shall be designed for assembly restraint against forces of 8g forward, 8g backward, 4.5g upward, and 4.5g lateral.

3.6.12 Fittings, connections, and miscellaneous openings.

3.6.12.1 Manual drain valve. A 0.750 inch fitting located at the bottom of the vessel, 0.750 inch pipe, and a 0.750 inch ball valve shall be provided. The valve shall be located to permit ready access and shall not discharge liquid on the assembly. The valve shall be capable of draining all water, fuel and sediment from the filter separator by gravity. The valve shall be installed below the sump of the housing.

3.6.12.2 Filter separator and vessel inlet connector. A 4 inch female, Class 1, quick disconnect inlet in accordance with A-A-59326 shall be provided. The quick disconnect shall be provided with a dust plug attached by means of a MIL-DTL-83420, Type II, 0.0625 inch diameter wire rope. The connector shall be horizontal, within the envelope dimensions of the base, and located to permit easy attachment of the 4 inch hose. The connect shall prevent the hose from rubbing against sharp corners or edges, which could cause damage to the hose.

3.6.12.3 Vessel outlet connector. A 4 inch, grooved type coupling in accordance with A-A-52592 and A-A-52598 shall be provided for connection to the flow-limiter water slug valve as specified in 3.6.8.

3.6.12.4 Pressure relief valve. A 0.750 inch angle pattern pressure relief valve shall be provided on top of each vessel. The pressure relief valve shall perform at any ambient temperature from +125°F with full impact of solar radiation of 360 British Thermal Units per square foot per hour (BTU/sq. ft. /hr.) minimum 4 hours, to -25°F (-32°C) minimum 3 days without benefit of solar radiation. The relief valve setting shall be rated at 110% of the rated operating pressure (see 4.5.1).

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3.6.12.5 Automatic air eliminator. An automatic air eliminator shall be installed on the highest point of the vessel and shall have a check valve feature. An air eliminator shall be provided with an orifice maximum 0.1250 inch, capable of operating at the pressure rating of the vessel. The liquid fuel and the liquid fuel formed from the mist shall be returned to the filter separator through the vapor trap inlet connection during the draining or defueling operation. The air eliminator shall be designed so that while under a fuel pressure of 0.250 and 150 psig (pounds per square inch gauge) it will discharge introduced air without discharge of liquid fuel. The air eliminator shall not leak when the vessel is completely filled with fuel and pressure is raised to 150 psig and shall be capable of performing in ambient temperatures ranging from -65° to +130°F. An air eliminator drainpipe to the outlet flow shall also be provided in order to drain the sump tank.

3.6.12.6 Sampling connections. Sampling connections shall be provided at the inlet and outlet connections to the housing. Each sampling connection shall consist of the following: 0.250 inch ball valve, 0.250 inch quick disconnect coupling and dust cap. The sampling connections shall be capable of accepting a sampling kit for drawing the samples required to assure fuel quality.

3.6.13 Spider assembly. Each filter separator shall contain a spider assembly to hold the coalescers and separators in position, to support them firmly against vibration. The method of stabilization shall assure an electrical bond between the spider and the vessel.

3.6.14 Coalescer and separator cartridges. Each filter separator shall be provided with coalescers and separators that have been qualified to the performance requirements of API Bulletin 1581, for Group II, Class B equipment.

3.6.15 Flow limiter water slug valve. A flow limiter water slug valve shall be provided which will limit the maximum flow to 660 +10 gpm and stop the fuel flow when predetermined water level is reached in the vessel and then permit fuel flow when the water level is lowered to a predetermined level. The pressure loss across the valve shall not exceed 10.4 psi during flow of 0.76 specific gravity grade JP-4 fuel in accordance with MIL-DTL-5624 at a flow rate of 600 gpm ±10 gpm. The float actuated pilot valve shall be located in the vessel.

3.6.16 Differential pressure gauge. The housing shall be equipped with a direct reading, piston type, differential pressure gauge that measures the differential pressure across both coalescers and separators. The gauge shall consist of a spring supported, corrosion resistant piston moving inside a glass cylinder, with high pressure applied on top of the piston and low pressure applied below it. Under a differential pressure of 30 psi, leakage past the piston shall not exceed 120 drops per minute. The cylinder shall have end flanges with O-ring seals. The high-pressure inlet of the gauge shall have a 10 micron pleated paper filter and the low pressure of 300 psi with a cylinder burst pressure minimum 1200 psi. Differential pressure range of the gauge through approximately 3 inches of piston movement shall be 0-30 psi with an accuracy of +0.5 psi, calibrated linearly with 1.0 psi scale graduations. High and low pressure connections shall be 0.2500 inch NPT female with a bar stock valve at each connection. Construction of the gauge shall be such that a 3 valve manifold is not necessary. If only one bar stock valve is closed, the gauge shall not be damaged by up to 3000 psi differential pressure in either direction. The differential pressure gauge shall be attached to the filter separator by a gauge panel.

3.6.17 Filter separator outlet connector. The quick disconnect outlet connector shall be provided with a dust plug and a 0.0625 inch diameter security wire rope that is nylon jacketed. The outlet connector shall be located downstream of the flow limiter water slug valve, shall be horizontal within the envelope dimensions of the base (see 3.6.3), and located to permit each attachment of the 4 inch hose. The connector shall prevent the hose from rubbing against sharp corners or edges, which could cause damage to the hose.

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3.6.18 Ground rod. A 5 foot ground rod shall be provided in accordance with A-A-55804 (Type II, style 1). The ground terminal shall be bolted to the base forming an electrical continuity with a maximum resistance of 0.5 ohm. Provisions shall be incorporated for storing and retaining the ground rod with the cable attached.

3.6.19 Sight gauge. A 0.50 inch armored, clear viewing liquid level gauge shall be provided for observing the water accumulation in the sump. The gauge shall be fabricated from a transparent material and contain a colored density-sensitive ball.

3.6.20 Cleaning. Components of the filter separator shall be cleaned prior to delivery to remove dirt; excess soldering; brazing and welding flux; welding slag; loose, spattered or excess solder; metal chips; and other foreign materials during and after assembly.

3.6.21 Finishing. The unit shall be corrosion resistant to weathering elements and be primed and finished in accordance with best commercial practices. The color shall be Forest Green, No. 24052, as designated in the FED-STD-595.

3.7 Operating environment requirements. The filter separator shall operate under the following environmental conditions without degradation in performance:

3.7.1 Operating temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of -40° to +125°F and after tested in accordance with 4.5.4.1.

3.7.2 Humidity. The equipment shall meet its performance and accuracy requirements in an operating environment of less than 98% relative humidity and tested in accordance with 4.5.4.2.

3.7.3 Air transportability test. The filter separator shall withstand transport loads outlined in 4.5.4.3 and 3.7.3.1.

3.7.3.1 Restraint criteria. Cargo items in their shipping configuration shall be capable of being restrained to and withstanding, as follows:

a. Without the loss of serviceability, the forces applied statically and independently, see Table I.

TABLE I. Applied forces

Direction	Load Factor
Up	2.0G (3.7G for nuclear cargo)
Down	4.5G

b. Without loss of structural integrity, the acceleration forces applied statically and independently. , see Table II.

TABLE II. Acceleration forces

Direction	Load Factor
Forward	3.0G
Aft	1.5G
Lateral	1.5G

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3.7.4 Fungus. The filter separator shall meet its performance and accuracy requirements in a fungus environment as encountered in tropical climates (see 4.5.4.4).

3.7.5 Salt and moisture test. The filter separator shall operate in moisture and salt-laden atmospheres (see 4.5.4.5 and 4.5.4.7).

3.7.6 Sand and dust test. The filter separator shall operate after exposure to sand and a dust-laden atmosphere (see 4.5.4.6).

3.8 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promote economically advantageous life cycle cost.

3.9 Identification plates. Equipment, assemblies, accessories, and parts shall be marked for identification in accordance with MIL-STD-130. The identification plates shall be permanently attached to the housing and, in addition to the usual MIL-STD-130 requirements, shall include the following information:

3.9.1 Mainframe identification plate. The mainframe data plate for the filter separator shall be permanently and legibly marked with the following information:

a. Manufacturer’s name:

b. Commercial and Government Entity (CAGE) Code:

c. Part identifying number:

d. Date of manufacturer:

e. Serial number (if different than part number):

f. Design flow rate:

g. Design minimum and maximum working pressure:

h. Specification number:

i. National Stock Number (NSN) if available:

3.9.2 Transportation identification plate. The transportation data plate, attached to the vessel near the identification plate and instructional panel, shall state:

a. Shipping weight (drained condition):

b. Loading cubage:

c. Overall height, width, and length:

d. Center-of-gravity horizontal and vertical location:

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection. (see 4.3).

4.2 First article inspection. First article inspections shall be performed on the sample as specified below:

4.2.1 Test samples. The first article test sample shall consist of one filter separator fabricated in the same facilities, using the same materials, components and production technologies that will be used for the production lot of the filter separators.

4.2.2 First article inspection tests. The first article inspection and tests shall consist of all the tests described under paragraphs 4.4 and 4.5.

4.3 Conformance inspection. The conformance inspection and tests shall consist of individual tests. The conformance inspection shall be conducted on 1 out of every 20 production quantity units. The filter separator shall be subjected to the tests as described under 4.5.

4.3.1 Examination of product (see 4.5.1).

4.3.2 Inspection and testing (see 4.5.2, 4.5.3, and 4.5.4).

4.4 Test methods.

4.4.1 Certification of Qualification. If this product has been previously tested and approved by the Government, submit certification of qualification under API Bulletin 1581, Group II, Class B. Include the description of qualification which contains element types and quantities and provide details of the configurations of vessels tested. Include the name of the Government Agency and the date of approval.

4.5 First article testing. Filter separators requiring first article tests shall be conducted in the presence of a contracting officer's designated Government representative. The contractor shall give the contracting officer 30 days notice prior to conductance of factory tests in order to schedule witnessing by the representative. The filter separator may be rejected if the stated requirements with respect to performance, design, configuration, materials, construction, workmanship, and markings are not met. The filter separator shall be subjected to the following tests:

4.5.1 Examination. The filter separator and its components shall be inspected to determine compliance with the requirements herein with respect to design, configuration, materials, construction, workmanship, markings and finish. First article examination shall be accomplished by means of a contractor-developed checklist listing the filter separator characteristics subject to examination, method of verification, and results.

4.5.2 Hydrostatic pressure test. The filter separator shall be subjected to a hydrostatic pressure of 225 psig per requirements of the ASME Unfired Pressure Vessel Code. This pressure, once attained, shall be maintained for a minimum period of 30 minutes while the pressure producing pump is disconnected

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or valved off. For conformance inspections, the time shall be reduced to 5 minutes. The initial and final pressures shall be recorded for the pre-production item and included in the test report. In addition, the inlet manifold or chamber (after installation) shall be blanked off and tested to 115 psig.

4.5.3 Full scale performance test. The filter separator with a full set of coalescer and separator elements shall be tested and meet the requirements of API Bulletin 1581 (Group II, Class B) at design flow rate of 600 gpm in accordance with API Bulletin 1581 Test Method Group II, Test Series No. 2, except as otherwise (see 4.5.4). The maximum flow through the filter separator shall be limited to 600 ±10 gpm. In addition, air introduced into the vessel shall be eliminated out through the air eliminator. A pressure loss of 20 psi shall not be exceeded from the inlet to the outlet connection at a flow rate of 600 gpm using 0.76 specific gravity grade JP-4 or JP-5/JP-8 fuel in accordance with MIL-DTL-5624 when the elements are in new condition.

4.5.4 Inspection and testing. The inspection and testing of the first article filter separator shall be conducted on a full-scale test system in accordance with API Bulletin 1581 and as specified herein. The test sample shall consist of a complete filter separator with elements installed. Samples shall be representative of a production lot.

4.5.4.1 Temperature tests. The filter separator shall meet the following temperature tests:

4.5.4.1.1 Low temperature test. Subject the filter separator to a temperature of -40°F, minimum 48 hours with relative humidity maximum 15%. Remove the filter separator from the test chamber and within 5 minutes after removal, operate the filter separator in accordance with the full performance test as specified in 4.5.3. The unit shall not experience any degradation in appearance or performance after testing.

4.5.4.1.2 High temperature test. Subject the filter separator to a temperature of +125°F, minimum 48 hours with a relative humidity maximum 15%. Remove the filter separator from the test chamber and within 5 minutes after removal, operate the filter separator in accordance with the full performance test as specified in 4.5.3. The unit shall not experience any degradation in appearance or performance after testing.

4.5.4.2 Humidity. The filter separator shall be subject to natural cyclic high humidity conditions that randomly varies the temperature between 79° and 95°F with the relative humidity reaching between 85 and 95% over a 24-hour period. Immediately following the humidity cycling, without drying or allowing the unit to dry, perform the full performance test as specified in 4.5.3. Unit shall not experience any degradation in appearance or performance.

4.5.4.3 Altitude/air transportability test. The filter separator unit shall be constructed so that all components will be restrained from causing any external damage in the event that applied forces of Table I or the acceleration forces of Table II are encountered. A detailed engineering evaluation with calculations and stress analysis on the components of the filter separator may be submitted in lieu of actual testing. Perform the full performance test as specified in 4.5.3 with no degradation in appearance or performance if actual testing is conducted.

4.5.4.4 Fungus resistance testing. The filter separator shall be subject to fungi and possibly abundant growth colonization of 50% or more of the area of the control item after 14 and 28 days. The unit shall not be cleaned for the first article for 72 hours prior to the fungus test. The unit shall not be contaminated before or during the test by handling. Perform the full performance test as specified in 4.5.3 with no degradation in appearance or performance.

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4.5.4.5 Salt-fog. Using spray apparatus, piping and collector reservoir materials that are non-reactive to salt spray, subject the filter separator to a continuously atomized, finely divided, wet, dense salt spray mixture (5% sodium chloride and 95% water) at a constant 95°F with minimum air circulation. Do not use immersion heaters in the exposure area. Uniformly distribute the salt fog over the filter generator at a fallout rate between 0.01 and 0.1 ounces/ $12in^2$ /hour minimum 48 hours. After the test, store the unit at ambient atmosphere minimum 48 hours for drying. Perform the full performance test as specified in 4.5.3 with no degradation in appearance or performance.

4.5.4.6 Sand and dust. The filter separator shall be subject to a test chamber temperature minimum 73°F with a relative humidity (maintained throughout the test) maximum 30%. Subject the filter separator to a blowing dust concentration (97 to 99 % by weight) of silicon dioxide with particle size ranging from 100 to 325 microns at air velocities ranging from 300 to 1750 feet/minute for minimum 6 hours. Stop the test and allow the test article to return to ambient conditions. Remove accumulated dust from the test article by brushing, wiping, or shaking. Do not remove dust by either air blast or vacuuming. Take care not to introduce additional dust into the test article. Perform the full performance test as specified in 4.5.3 with no degradation in appearance or performance. Repeat the test using sand concentrations of 2.2 to ±0.5 g/m. Perform the full performance test as specified in 4.5.3 with no degradation in appearance or performance.

4.5.4.7. Rain. The filter separator shall be subject to an instantaneous rainfall rate equivalent to 0.80 mm/minute, a minimum rate of 10 cm/hr. The rainfall intensities shall be accompanied by winds of 18 m/s (40 miles per hour). The rain shall be produced by a water distribution device of such design that the water is emitted in the form of droplets having a diameter range between 0.5 and 4.5 millimeters. The rain shall be dispersed completely over the test item for 15 minutes. The wind at the velocity specified in the test plan shall be maintained for minimum 30 minutes. Rotate the test item to expose the rain source to the other side of the test item that could be exposed to blown rain in the deployment cycle. Examine the test item and conduct a visual inspection. The unit shall allow any amount of free water to penetrate the test item. Measure and document the volume of water (if any) inside the protected areas of the filter separator.

4.5.5 Deviations from API Bulletin 1581. The following are filter separator requirements, which are deviations to API Bulletin 1581 for test requirements of Group II, Class B, Test Series 1, 2, and 3:

4.5.5.1 Fuel contamination limit. The allowable effluent fuel contamination limit for free water shall not exceed 10 parts per million and any single sample shall not exceed 15 parts per million as outlined in API Bulletin 1581.

4.5.5.2 Additive I. Additive I shall be Stadis 450 or similar in lieu of ASA 3.

4.5.5.3 Additive II. Additive II shall be HITEC-580 or similar conforming to MIL-PRF-25017 in lieu of HITEC E-515.

4.5.6 Single element test. Test series 3 shall be run in an appropriate scale element test vessel per API Bulletin 1581 and as modified by 4.5.5.

4.5.7 Coalescer structural test. A minimum of two coalescer elements, after being subjected to a full-scale performance test as described in 4.5.3, shall be subjected to a differential pressure test. Each element shall be capable of withstanding a differential pressure minimum 75 psi without bypassing of seals.

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4.5.8 Disassembly inspection. Upon completion of the tests specified above in 4.5.7, the filter separator shall be disassembled and inspected to determine the condition of the coalescer and separator elements. Defects in the element such as swelling of the elements, damaged elements, or damaged gasket shall be noted. Swelling of, or damage to, the elements or other parts shall be cause for rejection of the filter separator.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The filter separators described in this document are military unique. The FFU-15/E filter separator is intended for use with the A/E32R-14 fuel-servicing unit. When used with this aircraft refueling equipment, the fuel delivered to the fuel storage tanks is clean (no more than 1.0 mg/ liter solid contaminant) and dry (no more than 15 mg/liter undissolved water). Media migration of 10 fibers/liter. Usage will be primarily in rear combat areas and in remote environments requiring the unit to be skid mounted, air transportable, and be provided with tie downs, grounding rods, and finished in accordance with Air Force requirements. In addition, the unit must be made from lightweight materials resistant to specified environmental conditions without degradation in performance.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Document title, number, and date.

b. Issue of DoDISS to be cited in the solicitation and, if required the specific issue of referenced documents.

c. When examination, first article sampling and testing is required (see 4.2)

d. Number of product samples required (see 4.3)

e. Packaging requirements (see 5.1)

6.3 Part or Identifying Number (PIN). The PIN to be used for Fuel/Separators acquired to this specification is created as follows:

M	83041
Prefix for military specification	Specification number

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6.4 Subject term (key word) listing.

Air eliminator
Coalescer
Differential pressure gauge
Ground rod
Spider assembly
Water slug valve

6.5 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified in their corresponding inch/pound units. A request should be made to the contracting officer to determine if the product is acceptable. This is only if the manufacturer elects to use metric dimensions, as the basis upon which they derive or base a mechanical design feature. The contracting officer has the option of accepting or rejecting the product. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

Custodians:
Air Force - 99
Army - AT
DLA - IS

Preparing Activity:
Air Force - 82

Agent:
Air Force - 99

Project 4930-0061

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-83041C	**2. DOCUMENT DATE (YYYYMMDD)** 20000113

3. DOCUMENT TITLE **FILTER-SEPARATOR, LIQUID FUEL, SKID-MOUNTED, 600 GALLONS PER MINUTE**

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME ESTER BADILLO	b. TELEPHONE *Include Area Code)* (1) Commercial (216) 925-6314 (2) AUTOVON 945-6314
c. ADDRESS *(Include Zip Code)* SA-ALC/LDEE, BLDG. 171, 485 QUENTIN ROOSEVELT RD. KELLY AFB, TX 78241-6425	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman road, Suite 2533, Ft. Belvoir, VA 22060-2533 Telephone (703) 767-6888 AUTOVON 427-6888

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